

Package ‘rENM.ai’

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Type Package

Title GenAI tools for the rENM framework

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Description The rENM framework is a modular sysem for reconstucting and analyzing long-term ecological niche dynamics using time-indexed environmental data and species occurrence records. Its packages support climate-based niche modeling, temporal analysis of climatic suitability, and reproducible workflows for studying ecological change.

Depends R (>= 4.1.0)

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URL <https://github.com/rENM-Framework/rENM.ai>

BugReports <https://github.com/rENM-Framework/rENM.ai/issues>

Imports rENM.core,
curl,
httr2

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assemble_suitability_ai_package

Assemble GenAI suitability package for a species

Description

Collects suitability-trend rasters, summary tables, and species metadata for a single species, then builds an AI-ready package together with a prompt template and log entry. The function locates the project root via `rENM_project_dir()` and no longer relies on hard-coded paths.

Usage

```
assemble_suitability_ai_package(alpha_code)
```

Arguments

`alpha_code` Character. Four-letter species alpha code (e.g., "CASP").

Details

This function is part of the rENM framework's processing pipeline and operates within the project directory structure defined by `rENM_project_dir()`.

Pipeline context

- Copies pre-computed suitability rasters and tables into `suitability_package/`.
- Copies `\_species.csv` to a species-specific file and includes it in the package.
- Creates a `.zip` archive of the package contents.
- Copies the ChatGPT prompt template resolved via `system.file("chatgpt", "suitability_prompt.txt", package = "rENM.ai")` into the species `chatgpt/` directory.
- Appends a standardized processing-summary block to `\_log.txt`.
- Eliminates hard-coded home-directory paths by relying on `rENM_project_dir()`.

Directory layout

- `<project_root>/runs/<alpha_code>/Trends/suitability/`
- `<project_root>/runs/<alpha_code>/Summaries/chatgpt/`
- `<project_root>/data/_species.csv`

Expected inputs

- `<project_root>/runs/<alpha_code>/Trends/suitability/ <alpha_code>-Suitability-Change-Trend.tif`
- `<project_root>/runs/<alpha_code>/Trends/suitability/ <alpha_code>-Suitability-Trend-GAP-Range.tif`
- `<project_root>/runs/<alpha_code>/Trends/suitability/ <alpha_code>-Suitability-Trend-Percentage.tif`
- `<project_root>/runs/<alpha_code>/Trends/suitability/ <alpha_code>-Suitability-Trend-State-Annual.tif`
- `<project_root>/runs/<alpha_code>/Trends/suitability/ <alpha_code>-Suitability-Trend-State-Annual.tif`
- `<project_root>/runs/<alpha_code>/Trends/suitability/ <alpha_code>-Suitability-Trend-State-Annual.tif`
- `<project_root>/runs/<alpha_code>/Trends/suitability/ <alpha_code>-Suitability-Trend.tif`
- `<project_root>/data/_species.csv`

The shared species table is copied to <project_root>/runs/<alpha_code>/Trends/suitability/<alpha_code>-Species-Information.csv and included in the package.

Outputs

- Packaged files in <project_root>/runs/<alpha_code>/Summaries/chatgpt/ suitability_package/
- Zip archive <project_root>/runs/<alpha_code>/Summaries/chatgpt/ suitability_package.zip
- Prompt template copied beside the archive
- Updated species _log.txt

Missing inputs are reported with warning(). The zip archive is only created if at least one file is successfully copied.

Value

Named List (invisible).

- alpha_code – input species code
- pkg_dir – path to suitability_package
- alpha_chatgpt_dir – path to species chatgpt folder
- zip_path – path to created archive
- prompt_path – path to copied prompt file
- species_info_trends – trends-level species table
- species_info_pkg – table inside the package
- log_file – updated log file

Examples

```
## Not run:
assemble_suitability_ai_package("CASP")

## End(Not run)
```

render_ai_docs	<i>Render suitability-trend DOCX report to PDF and record processing summary</i>
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Description

Converts a species-specific DOCX report into PDF and appends a processing summary to the run log.

Usage

```
render_ai_docs(alpha_code, verbose = TRUE)
```

Arguments

alpha_code	Character. Species alpha code, for example "CASP".
verbose	Logical. If TRUE (default), prints diagnostic messages.

Details

This function is part of the rENM framework's processing pipeline and operates within the project directory structure defined by `rENM_project_dir()`.

Inputs

- One DOCX file per species located at `<rENM_project_dir()>/runs/<alpha_code>/Summaries/pages/<alpha_code>-Suitability-Trend-Analysis.docx`

Outputs

- A PDF written to the same directory: `<alpha_code>-Suitability-Trend-Analysis.pdf`
- A multi-line processing summary appended to `<rENM_project_dir()>/runs/<alpha_code>/_log.txt`

Methods

- Conversion uses LibreOffice soffice in headless mode.
- File sizes and time stamps are recorded for auditing.

Value

Character. Invisibly returns the path to the generated PDF.

Side effects:

- Writes a PDF file alongside the source DOCX.
- Appends a formatted record to the species-level `_log.txt`.

Examples

```
## Not run:
# Convert the Cassin's Sparrow report and view the resulting PDF path
pdf_path <- render_ai_docs("CASP")

## End(Not run)
```

```
submit_suitability_ai_package
```

Submit GenAI-ready climatic suitability trend package for a species

Description

Runs an end-to-end climatic suitability trend analysis for a single species by sending a zipped data bundle and custom prompt to the OpenAI Responses API, retrieving DOCX and PDF reports, and logging processing metadata and cost information.

Usage

```
submit_suitability_ai_package(
  alpha_code,
  model = "gpt-5.1",
  api_key = Sys.getenv("OPENAI_API_KEY")
)
```

Arguments

alpha_code	Character. Four-letter species alpha code (e.g., "CASP").
model	Character. OpenAI model identifier that supports the Responses API and code-interpreter tool. Default is "gpt-5.1".
api_key	Character. OpenAI API key. Defaults to Sys.getenv("OPENAI_API_KEY").

Details

This function is part of the rENM framework's processing pipeline and operates within the project directory structure defined by `rENM_project_dir()`.

Pipeline context

- Reads a species-specific prompt file.
- Uploads a zipped bundle of GeoTIFF and CSV inputs.
- Calls the Responses API with the code-interpreter tool enabled.
- Downloads a Word report and a PDF report created inside the container.
- Computes token usage and approximate OpenAI cost.
- Appends an eBird-style summary line to `_log.txt`.

Expected inputs

- Prompt file containing the placeholder string `<alpha_code>`.
- Zipped bundle of raster and tabular data required by the prompt.

Authentication

- Uses `api_key`, defaulting to `Sys.getenv("OPENAI_API_KEY")`.
- Leading and trailing whitespace are removed from the API key before use.
- If present, `OPENAI_ORG_ID` and `OPENAI_PROJECT_ID` are added as request headers to support explicit organization or project routing.

Directory layout under `rENM_project_dir()`

```
<project_dir>/runs/CASP/Summaries/chatgpt/suitability_package.zip
<project_dir>/runs/CASP/Summaries/chatgpt/suitability_prompt.txt
<project_dir>/runs/<alpha_code>/Summaries/pages/
  <alpha_code>-Suitability-Trend-Analysis.docx
<project_dir>/runs/<alpha_code>/Summaries/pages/
  <alpha_code>-Suitability-Trend-Analysis.pdf
<project_dir>/runs/<alpha_code>/_log.txt
```

Original layout reference

```
~/rENM/runs/CASP/Summaries/chatgpt/suitability_package.zip
~/rENM/runs/CASP/Summaries/chatgpt/suitability_prompt.txt
~/rENM/runs/<alpha_code>/Summaries/pages/
  <alpha_code>-Suitability-Trend-Analysis.docx
~/rENM/runs/<alpha_code>/Summaries/pages/
  <alpha_code>-Suitability-Trend-Analysis.pdf
~/rENM/runs/<alpha_code>/_log.txt
```

Outputs

- DOCX and PDF reports saved to the pages directory.
- Updated _log.txt containing a processing summary.

Cost model

- Input tokens: USD 1.25 per 1M tokens.
- Output tokens: USD 10.00 per 1M tokens.
- Code-interpreter session: USD 0.03 flat.

Value

List (invisible) with components

- response - parsed JSON response from the API.
- docx_path - file path of the DOCX report, or NA if not found.
- pdf_path - file path of the PDF report, or NA if not found.
- elapsed_sec - total elapsed time in seconds.
- input_tokens, output_tokens, total_tokens - approximate token counts.
- est_total_cost - approximate cost in USD.

Examples

```
## Not run:  
res <- submit_suitability_ai_package("CASP")  
res$docx_path  
res$pdf_path  
  
## End(Not run)
```

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